

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 05

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1 The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

- A 3
- B 1
- C -1
- D -3
- E 5
- F 2

- 2 The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?
- A** The estimate is $\frac{8}{3}$ and it is exact.
- B** The estimate is 3 and it is an overestimate.
- C** The estimate is $\frac{5}{2}$ and it is an overestimate.
- D** The estimate is $\frac{5}{2}$ and it is an underestimate.
- E** The estimate is 3 and it is an underestimate.

3 Convert 135° to radians.

- A** $\frac{4\pi}{3}$
- B** $\frac{3\pi}{8}$
- C** $\frac{4\pi}{3}$
- D** $\frac{5\pi}{4}$
- E** $\frac{6\pi}{2}$
- F** $\frac{3\pi}{5}$

- 4 Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?
- A 51
 - B 2
 - C 100
 - D 99
 - E 52
 - F 50

- 5 In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?
- A 36
 - B 35
 - C 19
 - D 37
 - E 64
 - F 18

6 The statement '**if a triangle is equilateral then it is isosceles**' is true. Which of the following must therefore also be true?

- I** If a triangle is isosceles then it is equilateral.
- II** If a triangle is not isosceles then it is not equilateral.
- III** If a triangle is not equilateral then it is not isosceles.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

7 Which of the following is the negation of $x > 3$ and $y \leq 5$?

A $x < 3$ and $y > 5$

B $x < 3$ or $y \geq 5$

C $x \leq 3$ or $y \geq 5$

D $x \leq 3$ and $y > 5$

E $x \leq 3$ or $y > 5$

- 8** Consider the claim: **for every positive integer n , $n^2 + n$ is even.** Which of the following is a complete and valid proof?
- A** $n^2 + n$ is even because both n^2 and n are even.
- B** If n is even then n^2 is even and so $n^2 + n$ is even.
- C** Suppose $n^2 + n$ is odd. Then n is odd, so n^2 is odd, so $n^2 + n$ is even, which is a contradiction... unless n is even.
- D** $1^2 + 1 = 2$, $2^2 + 2 = 6$, $3^2 + 3 = 12$, which are all even, so the claim holds.
- E** $n^2 + n = n(n + 1)$ is the product of two consecutive integers, one of which must be even, so the product is even.

- 9 x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 10** Which of the following statements is/are true?
- I** Every rhombus is a parallelogram.
 - II** Every rectangle is a rhombus.
 - III** The diagonals of a kite are perpendicular.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

11 Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

A $x = -1, y = -1$

B $x = 1, y = 2$

C $x = 2, y = -2$

D $x = 0, y = 0$

E $x = 2, y = 2$

12 Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I** For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II** For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III** For every real k the equation has at least one solution.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

13 a and b are positive real numbers. Which of the following statements is/are true?

I If $a^2 = b^2$ then $a = b$.

II If $2^a = 2^b$ then $a = b$.

III If $a^b = 1$ then $a = 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 14** A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?
- A** The n th line starts at n^2 and has $n + 1$ numbers on the left and n on the right.
- B** The n th line starts at $n(n + 1)$ and has $n + 1$ numbers on the left.
- C** The n th line starts at n^2 and has n numbers on the left and $n + 1$ on the right.
- D** The n th line starts at n^2 and has n numbers on each side.
- E** The n th line starts at $2n - 1$ and has $n + 1$ numbers on the left.

15 Which of the following statements can be disproved by a single counterexample?

- I** Every prime number greater than 2 is odd.
- II** Every integer of the form $4k + 3$ is prime.
- III** There exists an even prime number.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 16** Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 17** A bag contains red and blue balls only. If one red ball is removed, one quarter of the remaining balls are red. If instead two blue balls are removed, one third of the remaining balls are red. How many balls are in the bag?
- A** 16
- B** 20
- C** 14
- D** 8
- E** 17
- F** 12

- 18** A student solves the inequality $\frac{x+1}{x-2} < 3$.
- I** Multiply both sides by $(x-2)$: $x+1 < 3x-6$.
- II** So $7 < 2x$.
- III** So $x > \frac{7}{2}$.

Which of the following is correct?

- A** The first error is on line I: multiplying by $x-2$ is only valid as written when $x-2 > 0$.
- B** The solution is completely correct.
- C** The first error is on line II: it should be $7 < 4x$.
- D** The first error is on line III: it should be $x < \frac{7}{2}$.
- E** The first error is on line I: you cannot multiply an inequality by an expression.

- 19** A sequence has $u_1 = 3$ and $u_{n+1} = \frac{u_n + 5}{2}$. Which of the following statements is/are true?
- I** The sequence is increasing.
 - II** $u_n < 5$ for all n .
 - III** $u_n = 5 - 2^{2-n}$ for all n .
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

20 f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?

I $f(x) = 0$ for some x with $0 \leq x \leq 4$.

II The area between the curve and the x -axis from 0 to 4 is 0.

III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST